

The 'sopra-base' class

Documentation for the 'sopra-base' class | Version v1.0.01

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1 General

1.1 Why, wherefore, what for?

This $\text{\LaTeX}$ 2_ε document class was written as part of the Sopra in the 2019 winter semester and 2020 summer semester and serves as the foundation for the look of all documents of *Team 20*, and may still be fundamentally changed over the course of its development. The design so far is based on the assignment template of Ulm University, whereby an interface was created that should not require changes to the $\text{\LaTeX}$ code even for varying visual properties of the document.

To visualize the individual code excerpts, the `sopra-listings` package is used. It is not relevant for the use of the document class itself.

The associated class should also be embedded in this document: .

1.2 Dependencies

This package is based on the `article` class,⁽¹⁾ which is included with every $\text{\LaTeX}$ distribution, as well as the following packages (all passed arguments are also given, whereby a star is set in case a value is passed):

- | | | |
|---|---------------------------------|--|
| ▪ <code>tikz</code> | ▪ <code>sfmath</code> | ▪ <code>enumitem</code> |
| ▪ <code>fontenc</code> ^[T1] | ▪ <code>lmodern</code> | |
| ▪ <code>inputenc</code> ^[utf8] | ▪ <code>microtype</code> | ▪ <code>nowidow</code> ^[all] |
| ▪ <code>babel</code> ^[ngerman,main=english] | ▪ <code>scrlayer-scrpage</code> | |
| ▪ <code>geometry</code> ^[margin*, a4paper,bottom*] | ▪ <code>titlesec</code> | ▪ <code>hyperref</code> ^[hidelinks] |

All of these packages should be part of the common $\text{\LaTeX}$ distribution (in addition, the TikZ library `calc` is used, as well as `shapes` if a group number is given via `\groupnum`^{→ p. 3} or `\setTeam`^{→ p. 4}).

1.3 Installation

The class is not shipped as a `.dtx`, which is why the following options are available:

- The document class can be placed in the same directory as the document. In this case, the inclusion instruction is:

```
\documentclass{sopra-base}
```

- The document class can be placed in a subdirectory/in a directory shipped with the document. In this case, it is specified via the (relative) path:

```
\documentclass{./My/Path/to/sopra-base}
```

⁽¹⁾<https://ctan.org/pkg/article>

- One can place the class (via a symlink or similar) into one's own *texmf* tree. For example, on Linux using texlive, the class can be placed here: `~/texmf/tex/latex/`. The directory can be created and subsequently updated via `texhash ~/texmf`. Now the class can be used like any other installed class:

```
\documentclass{sopra-base}
```

1.4 Further peculiarities

In version v1.0.01 (`\thesobversion`^{→ p. 4}) there are no further peculiarities.

2 Class configuration

2.1 Accepted parameters

Like most, the document class accepts arguments. In addition to the arguments accepted by `article`, the following are listed below, whereby for arguments with a “counter” option the one active by default is written first and the other in brackets. So implicitly:

```
\documentclass[final,nopar,sffont,defaultmode,showmail]{sopra-base}
```

is invoked. Whereas with:

```
\documentclass[rmfont]{sopra-base}
```

we can have the document set in the well-known *Roman* font.

▷ `final` (draft)

In draft mode some elements such as graphics cannot be typeset, in order to speed up the compilation process. This option is likewise accepted by many other packages as well.

▷ `nopar` (dopar)

Using `dopar` we set the paragraph indentation to the default value set by $\text{\LaTeX}$.

▷ `sffont` (rmfont)

Using `rmfont` the font is set back to *Roman*.

▷ `defaultmode` (print)

Using `print`, colors are reduced, or in part completely removed (converted to grayscale), in order to achieve a better result in print.

▷ `showmail` (hidemail)

With `hidemail` the e-mail address is not shown when listing the authors.

2.2 Further styles

In the current version (v1.0.01) no alternative styles and variants are supported yet. They will presumably be realized via further parameters or subclasses.

3 Command overview

As a *base* class, the class does not add a large set of commands for the user. This task falls to the extending classes and packages.

3.1 Setting data

The following commands should be set in the preamble and also configure the metadata of the respective document (i.e. the document properties). They can be overwritten as often as desired (up to the point at which the document begins). *Developer note:* All of these fields are available via `\sob@register@⟨Name⟩`, or their 'short' variants via `\sob@register@short@⟨Name⟩`.

▷ `\title[short]{NewTitle}`

Sets the title of the document to NewTitle.

▷ `\subtitle[short]{NewTitle}`

Sets the subtitle of the document to NewTitle.

▷ `\brief[short]{ShortDescription}`

Sets the short description of the document to ShortDescription. This data field is not used in the base class.

▷ `\authors[short]{AuthorList}`

This can be used to set the list of authors. It is recommended to use `\addAuthor`^{→ p. 4} for this.

▷ `\duedate[short]{YYYY-MM-DD}`

Sets the date of the document. `\DateConvert`^{→ p. 4} is used for the conversion.

▷ `\supervisor[short]{Name}`

Sets the supervisor to Name. This data field is not used in the base class.

▷ `\keywords[short]{Keywords}`

Sets the keywords for the metadata to Keywords.

▷ `\group[short]{Name}`

Sets the group of the respective creator to Name.

▷ `\groupnum[short]{Number}`

In case of a numbering (or symbol, etc.) of the group, sets this to Number.

3.2 Helpful when setting data

- ▷ `\addAuthor{Name (Email)}`

Adds an author to the list of authors (`\authors`^{→ p. 3}). For example: `\addAuthor{Florian Sihler (florian.sihler@uni-ulm.de)}`

- ▷ `\setTeam{Name Number}`

Allows `\group`^{→ p. 3} and `\groupnum`^{→ p. 3} to be set in a single call. For example: `\setTeam{Affenbande 42}` or `\setTeam{{Mega Team} A}`.

3.3 Displaying the data

- ▷ `\typesetAuthors`

Typesets the authors according to showmail with or without an e-mail address, so `\typesetAuthors` yields: Florian Sihler (florian.sihler@uni-ulm.de). For multiple authors, commas and the “and” are set correctly.

- ▷ `\maketitle`

Sets the title based on the data set

- ▷ `\thesobversion`

Outputs the current version of the sopra-base document class. Note: via `\value{sobversion}` the version can be obtained as a 4-digit number: 1001.

- ▷ `\DateConvert{YYYY-MM-DD}`

Converts a date into a nicer form, like so:

`\DateConvert{2019-12-02}` % → 02. December 2019

3.4 General formatting commands

- ▷ `\imptext{Text}`

Highlights Text as *very important*.

- ▷ `\notetext{Text}`

Sets Text as an incidental note.

A Related classes and packages

In the course of this $\text{\LaTeX}2_{\epsilon}$ document class, a few other packages have also come into being. As of the current compilation state, the following exist (packages and classes without a link are included alongside this one and can be found here: <https://github.com/EagleoutIce/sopra-collection>):

- `eagle-maps`⁽²⁾
- `LILLY-Framework`⁽³⁾
- `sopra-documentation`
- `sopra-models`
- `sopra-listings`
- `sopra-requirements`
- `sopra-attachments`

⁽²⁾<https://github.com/EagleoutIce/eagle-maps>

⁽³⁾<https://github.com/EagleoutIce/LILLY>